José Manuel Rebollo Bernal
Task 07 DB
28/04/2026

Write the code used to perform the following tasks from the MongoDB shell:

1. Create a database called zoo.

```
test> use zoo
```

2. Insert a collection called animals.

```
zoo> db.createCollection("animals")
```

3.Confirm that the database and the collection have been successfully created.

```
zoo> show dbs
zoo> show collections
```

4. Insert the documents contained in the animals.json file.

```
zoo> db.animals.insertMany([

   {
     "_id": 1,
…
   {
     "_id": 70,
     "name": "Horn",
     "age": 7,
     "gender": "Male",
     "species": "Rhinoceros",
     "habitat": "Grasslands",
     "birthplace": "Kenya",
     "parents": [
       { "name": "Unknown", "gender": "Male" },
       { "name": "Unknown", "gender": "Female" }
     ],
     "caretakers": [{ "idCaretaker": 614, "nameCaretaker": "Nathan Hall" }]
   }
 ])
{
  acknowledged: true,
```

5- Write a MongoDB query to find animals that have Rainforest as their habitat and are attended by a caretaker named Emily Brown.

```
zoo> db.animals.find({
  $and: [
    { habitat: "Rainforest" },
    { "caretakers.nameCaretaker": "Emily Brown" }
  ]
})
```

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6- Write a MongoDB query to display only the name, species, and habitat of all animals sorted by their species in ascending order.

```
zoo> db.animals.find(
  {},
  { _id: 0, name: 1, species: 1, habitat: 1 }
).sort({ species: 1 })
```

7 .Write a MongoDB query to find animals with the species Tiger and display only their name and age, sorted by age in descending order.

```
zoo> db.animals.find(
   { species: "Tiger" },
   { _id: 0, name: 1, age: 1 }
 ).sort({ age: -1 })
```

8. Write a MongoDB query to insert the next animal into the animals collection:
name: "Leo",
age: 7,
gender: "Male",
species: "Lion",
habitat: "Savanna",
birthplace: "Masai Mara, Kenya",
parents: "Unknown", gender: "Male", Unknown", gender: "Female"
caretakers: id: 501, name: "Jake Adams" }]

```
zoo> db.animals.insertOne({
   name: "Leo",
   age: 7,
   gender: "Male",
   species: "Lion",
   habitat: "Savanna",
   birthplace: "Masai Mara, Kenya",
   parents: [
     { name: "Unknown", gender: "Male" },
     { name: "Unknown", gender: "Female" }
   ],
   caretakers: [
     { idCaretaker: 501, nameCaretaker: "Jake Adams" }
   ]
 })
{
  acknowledged: true,
  insertedId: ObjectId('69f0cae1b44f30ce3c3682d1')
}
```

9. Write a MongoDB query to update the age (to 8) of the animal you just inserted using

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the $set operator

```
zoo> db.animals.updateOne (
       { name: "Leo" },
       {$set: { age: 8 } }
        )
```

10. Write a MongoDB query to rename the habitat field to environment for all animals in the animals collection.

```
zoo> db.animals.updateMany (
       {},
       {$rename: {habitat: "environment"} }
       )
```

11- Write a MongoDB query using the aggregation framework to find animals aged less than 10 and sort them alphabetically by their name.

```
zoo> db.animals.aggregate ( [
       {$match: { age: {$lt: 10} } },
       { $sort: { name: 1} }
       ] )
```

12. Write a MongoDB query using the aggregation framework to group animals by their species and count the number of animals in each species.

```
zoo> db.animals.aggregate ( [
       {$group: { _id: "$species",
       count: { $sum: 1 } } }
       ] )
```

13. Write a MongoDB query to create a new collection called Caretakers by extracting all caretakers from the animals collection. Ensure that caretakers do not repeat and include an array of the animals they attend to.

```
zoo> db.animals.aggregate([
   { $unwind: "$caretakers" },
   { $group: {
       _id: "$caretakers.idCaretaker",
       nameCaretaker: { $first: "$caretakers.nameCaretaker" },
       animals: { $push: "$name" }
     }
   },
   { $project: {
       _id: 0,
       idCaretaker: "$_id",
       nameCaretaker: 1,
       animals: 1
     }
   },
   { $out: "Caretakers" }
 ])
```

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14. Write a MongoDB query to find the top 3 oldest animals.

```
zoo> db.animals.find()
       .sort
              ({ age: -1 })
       .limit(3)
```

15.Write a MongoDB query to display animals between the 6th and 12th entries in the collection based on their insertion order.

```
zoo> db.animals.find()
       .skip(5)
       .limit(7)
```

16. Write a MongoDB query using the aggregation framework to create a new collection called Habitats. This collection should group animals by their environment (formerly habitat) and include the following fields.

- The environment name.
- The total number of animals in that environment.
- An array of unique species in that environment, sorted alphabetically.
- An array of unique caretakers who attend animals in that environment.

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```
zoo> db.Habitats.find()
[
  {
    _id: ObjectId('679370346cee80a6809e83a7'),
    totalAnimals: 11,
    environment: 'Grasslands',
    species: [ 'Buffalo', 'Elephant', 'Kangaroo', 'Rhinoceros' ],
    caretakers: [
      'Alice Brown',
      'Alice Green',
      'Alice White',
      'Emily Brown',
      'Emily Green',
      'Jake Adams',
      'John Smith',
      'Mark Taylor',
      'Nathan Hall'
    ]
  },
  {
    _id: ObjectId('679370346cee80a6809e83a8'),
    totalAnimals: 10,
    environment: 'Savanna',
    species: [ 'Giraffe', 'Lion' ],
    caretakers: [
      'Alice Green',
      'Emily Brown',
      'Jake Adams',
      'John Smith',
      'Mark Taylor',
      'Michael Davis',
      'Sophia Green'
    ]
  },
  {
    _id: ObjectId('679370346cee80a6809e83a9'),
    totalAnimals: 3,
    environment: 'River',
    species: [ 'Hippopotamus' ],
    caretakers: [ 'Alice Green', 'Mark Brown', 'Sophia Taylor' ]
  },
```

```
zoo> db.animals.aggregate([
  { $unwind: "$caretakers" },
  { $group: {
      _id: "$environment",
      totalAnimals: { $sum: 1 },
      species: { $addToSet: "$species" },
      caretakers: { $addToSet: "$caretakers.nameCaretaker" }
    }
  },
  { $project: {
      _id: 0,
      environment: "$_id",
      totalAnimals: 1,
      species: 1,
      caretakers: 1
    }
  },
  { $out: "Habitats" }
])
```